Department of Computer Science	Online Final Exam for the course *Logic and Sets*
Vrije Universiteit Amsterdam	June 4, 2020, 18:30–22:00

(Students who only hand in solutions for part 2 because they want to reuse their mark for the midterm exam, have until 20:45.)

This exam has 9 pages and 8 exercises.

The result will be computed as (the total number of points plus 10) divided by 10.

Please motivate all answers!

By participating in this exam, I declare to understand that taking an online exam during this corona crisis is an emergency measure to prevent study delays as much as possible. I know that fraud control will be tightened and realize that a special appeal is being made to trust my integrity. With this statement, I promise to:

- make this exam completely on my own, and not share my solutions with other students,
- not consult sources other than explicitly mentioned in the examination, and
- make myself available for any oral explanation of my answers.

Part 1

1. Island puzzle (10 points)

On the island of liars and truth speakers, everybody is either a liar (who always lies) or a truth speaker (who always speaks the truth). You meet islanders A, B, and C.

A says: "Either B or C is a liar." (This is an exclusive or, i.e., $\oplus$.)

B says: "If C is a liar, then I am a truth speaker."

C says: "A is a truth speaker."

Determine, either through logical reasoning or via a truth table, who of A, B, and C is a liar and who is a truth speaker.

If you use logical reasoning, then at the end also check that the status of the islanders (liar or truth speaker) is in line with the truth values of their statements.

Solution: Logical reasoning approach: Suppose C is a truth speaker.

Then, by C's statement, A is also a truth speaker.

Then A's statement implies that B is a liar.

But since C is a truth speaker, B's statement is true (since C a truth speaker), contradicting that B is a liar.

So we conclude that C is a liar.

Then C's statement implies that A is a liar too.

Then A's statement implies that B is a liar too.

So we conclude that A, B and C are all liars.

This is in line with their three statements. Since B and C are both liars, A's statement is false. Since C and B are liars, B's statement is false. And since A is a liar, C's statement is false.

Truth table approach: Truth values for t_A, t_B and t_C need to be determined such that the three formulas $t_A \leftrightarrow (\neg t_B \oplus \neg t_C)$, $t_B \leftrightarrow (\neg t_C \rightarrow t_B)$ and $t_C \leftrightarrow t_A$ are all true.

t_A	t_B	t_C	$\neg t_A$	$\neg t_B$	$\neg t_C$	$\neg t_B \oplus \neg t_C$	$t_A \leftrightarrow (\neg t_B \oplus \neg t_C)$	$\neg t_C \rightarrow t_B$	$t_B \leftrightarrow (\neg t_C \rightarrow t_B)$	$t_C \leftrightarrow t_A$
T	T	T	F	F	F	F	F	T	T	T
T	T	F	F	F	T	T	T	T	T	F
T	F	T	F	T	F	T	T	T	F	T
T	F	F	F	T	T	F	F	F	T	F
F	T	T	T	F	F	F	T	T	T	F
F	T	F	T	F	T	T	F	T	T	T
F	F	T	T	T	F	T	F	T	F	F
F	F	F	T	T	T	F	T	F	T	T

Only in the last row do the 8th, 10th and 11th column all have a T. So $t_A = t_B = t_C = \mathtt{F}$, meaning that A, B and C are liars.

2. CNF and DPLL (4.5 + 8 points)

(a) Extract a formula ϕ in CNF from the following truth table:

p	q	ϕ
T	T	F
T	F	F
F	T	T
F	F	F

Solution: For each line in the truth table where ϕ is F, a conjunct is needed to express that the truth values on this line are not allowed.

For the first line, this is $\neg p \vee \neg q$.

For the second line, this is $\neg p \vee q$.

For the third line, this is $p \vee q$.

So the CNF extracted from the truth table is $(\neg p \vee \neg q) \wedge (\neg p \vee q) \wedge (p \vee q)$.

(b) Apply the DPLL procedure to this CNF to try to find a satisfaction of this formula. Why are you not surprised by the outcome?

Solution: There are no unit clauses or pure (positive or negative) propositional variables.

First substitute $\top$ for p, yielding $(\neg\top \vee \neg q) \wedge (\neg\top \vee q) \wedge (\top \vee q)$, which reduces to $(\perp \vee \neg q) \wedge (\perp \vee q) \wedge \top$ and then to $\neg q \wedge q$. Now both substituting $\top$ and substituting $\perp$ for q trivially produces the outcome $\perp$.

So we need to backtrack, and substitute $\perp$ for p, yielding $(\neg \perp \vee \neg q) \wedge (\neg \perp \vee q) \wedge (\perp \vee q)$, which reduces to $(\top \vee \neg q) \wedge (\top \vee q) \wedge q$ and then to $\top \wedge \top \wedge q$ and then to q. Now substituting $\top$ for q produces the outcome $\top$. So we have found a satisfaction of the formua in CNF: $p = \mathtt{F}$ and $q = \mathtt{T}$.

This is no surprise because this is exactly the line in the truth table where the original formula ϕ is $\mathtt{T}$.

3. Sets (5.5 + 5 points)

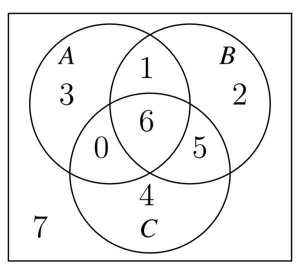
(a) Suppose that in a universe U with 28 elements, there are three sets A, B and C of which we know that

$$\#A = 10 \qquad \#B = 14 \qquad \#C = 15$$
$$\#(A \cap B) = 7 \qquad \#(B \cap C) = 11 \qquad \#(C \cap A) = 6$$
$$\#(A \cap B \cap C) = 6$$

Draw a Venn diagram of the sets A, B and C, determine the number of elements in every region of your Venn diagram, and use your Venn diagram to determine

$$\#(B \cup C) \qquad \text{and} \qquad \#(A \setminus (B \cap C))$$

Solution: The filled-in Venn diagram looks like this:



From the Venn diagram we read off that

$$\#(B \cup C) = 18, \qquad \#(A \setminus (B \cap C)) = 4.$$

(b) Using the rules of the algebra of sets, prove that the following set-theoretic equality holds (for all sets A, B, and C):

$$(B \setminus A)' \cup (C \setminus A)' = (B \cap C)' \cup A$$

Solution:

$$\begin{aligned}
(B \setminus A)' \cup (C \setminus A)' &= (B \cap A')' \cup (C \cap A')' && \text{(definition)} \\
&= (B' \cup A'') \cup (C' \cup A'') && \text{(De Morgan)} \\
&= (B' \cup A) \cup (C' \cup A) && \text{(involution)} \\
&= (B' \cup C') \cup (A \cup A) && \text{(reorder)} \\
&= (B' \cup C') \cup A && \text{(idempotence)} \\
&= (B \cap C)' \cup A && \text{(De Morgan)}
\end{aligned}$$

4. Ordering relations (8 + 2 + 2 points)
Consider the ordering relation *IsDivisorOf* on the set A defined by

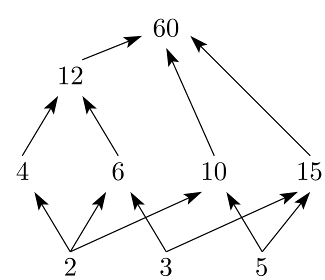
$$A := \{2, 3, 4, 5, 6, 10, 12, 15, 60\}.$$

(a) Use the algorithm that you have learned to construct the Hasse diagram that represents the *IsDivisorOf* ordering relation on A: explicitly write down the sets G_x and H_x (with $x \in A$) obtained in the construction, and then draw the Hasse diagram.

Solution: The sets G_x and H_x are as follows:

$$\begin{aligned}
G_2 &= \{4, 6, 10, 12, 60\} & H_2 &= \{4, 6, 10\} \\
G_3 &= \{6, 12, 15, 60\} & H_3 &= \{6, 15\} \\
G_4 &= \{12, 60\} & H_4 &= \{12\} \\
G_5 &= \{10, 15, 60\} & H_5 &= \{10, 15\} \\
G_6 &= \{12, 60\} & H_6 &= \{12\} \\
G_{10} &= \{60\} & H_{10} &= \{60\} \\
G_{12} &= \{60\} & H_{12} &= \{60\} \\
G_{15} &= \{60\} & H_{15} &= \{60\} \\
G_{60} &= \varnothing & H_{60} &= \varnothing
\end{aligned}$$

so this is the corresponding Hasse diagram:



(b) Does the set A have a smallest element according to the ordering relation? If so, please give this smallest element. If not, please list all the minimal elements of the set A.

Solution: There is no smallest element; the minimal elements are 2, 3 and 5.

(c) Does the set A have a largest element according to the ordering relation? If so, please give this largest element. If not, please list all the maximal elements of the set A.

Solution: There is a largest element, namely 60.

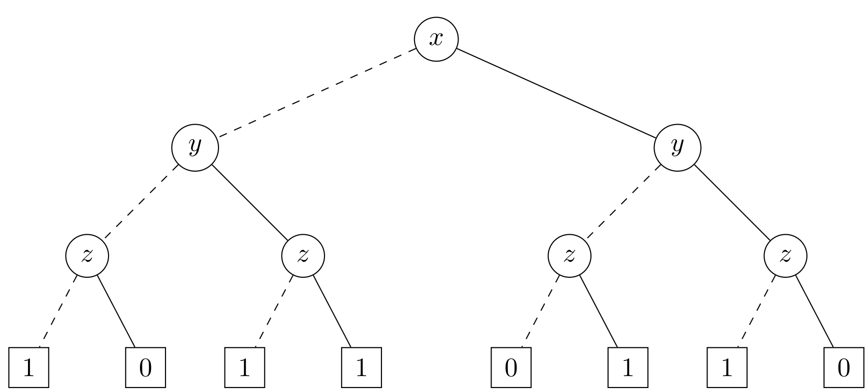
Part 2

5. OBDDs (2 + 4.5 + 6 points)
Let the propositional formula ϕ, with propositional variables x, y, z, have the following truth table:

x	y	z	ϕ
T	T	T	F
T	T	F	T
T	F	T	T
T	F	F	F
F	T	T	T
F	T	F	T
F	F	T	F
F	F	F	T

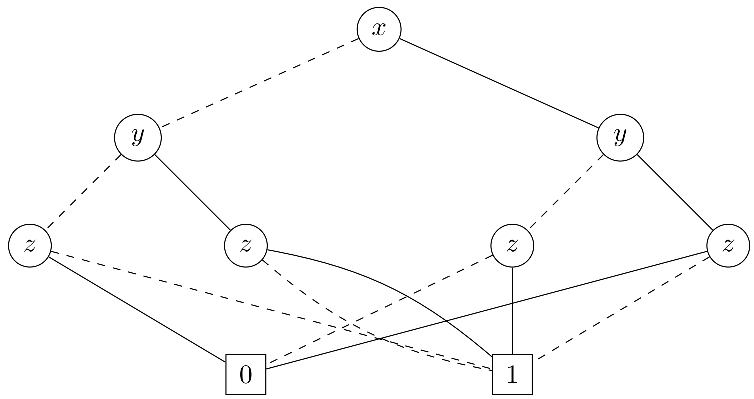
(a) Give the binary decision tree for ϕ, against the variable order x, y, z.

Solution:

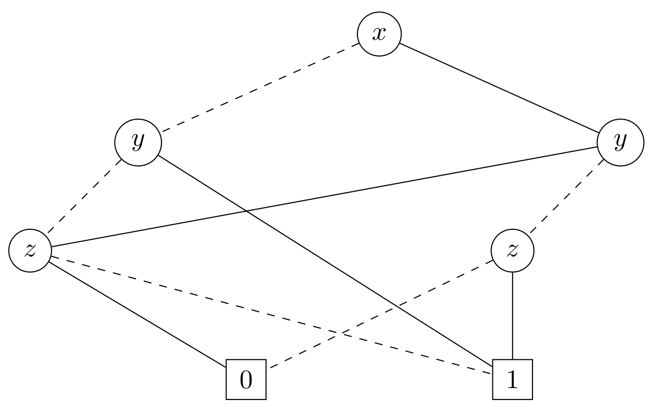


(b) Reduce this binary decision tree to a reduced OBDD. Clearly specify which reduction steps you apply.

Solution: Collapse leaves (C1).



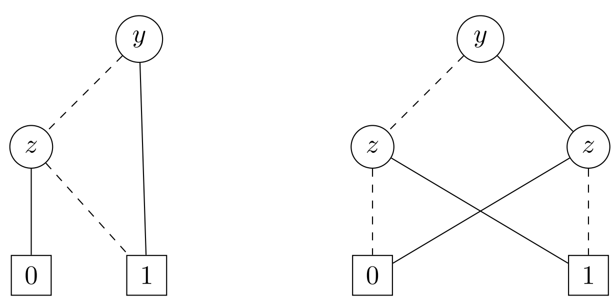
Eliminate node of which the 0- and 1-transition lead to the same node (C2).



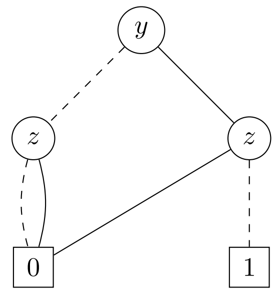
Now we have obtained the reduced OBDD.

(c) Construct an OBDD for $\forall x\, \phi$ from the reduced OBDD for ϕ, and reduce the resulting OBDD.

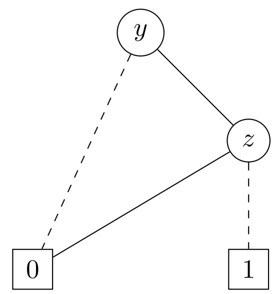
Solution: Taking $x = 0$ and $x = 1$, we get the following two OBDDs, respectively:



To construct the OBDD for $\forall x\, \phi$, we compute the conjunction of these two OBDDs:



Finally, to obtain a reduced OBDD, we apply (C2):



6. **Predicate logic** (5 + 5 points)
For each of the following pairs of formulas, either argue that they are semantically equivalent, or give a model on which their truth values are different. In case semantic equivalence does not hold, explain moreover whether one of the formulas semantically entails the other.

(a) $\neg(\exists x\, C(x) \vee \exists x\, D(x))$ and $\forall x\, (\neg C(x) \wedge \neg D(x))$

Solution: These formulas are semantically equivalent, which can be seen by the following derivation:

$$\begin{array}{rl} & \neg(\exists x\, C(x) \vee \exists x\, D(x)) \\ \equiv & \neg\exists x\, C(x) \wedge \neg\exists x\, D(x) \\ \equiv & \forall x\, \neg C(x) \wedge \forall x\, \neg D(x) \\ \equiv & \forall x\, (\neg C(x) \wedge \neg D(x)) \end{array}$$

(b) $\neg(\exists x\, C(x) \wedge \exists x\, D(x))$ and $\forall x\, (\neg C(x) \vee \neg D(x))$

Solution: These formulas are not semantically equivalent. Consider a model $M = \{a, b\}$ where C holds only for a while D holds only for b.

The first formula does not hold on this model, because $\exists x\, C(x)$ holds (since $C(a)$) and $\exists x\, D(x)$ holds (since $D(b)$). Hence $\exists x\, C(x) \wedge \exists x\, D(x)$ holds and so $\neg(\exists x\, C(x) \wedge \exists x\, D(x))$ does not hold on M.

The second formula does hold on this model, because $\neg D(a)$ and $\neg C(b)$, so for both elements x in the model, $\neg C(x) \vee \neg D(x)$ holds.

The first formula semantically entails the second formula. Note that $\neg(\exists x\, C(x) \wedge \exists x\, D(x)) \equiv \forall x\, \neg C(x) \vee \forall x\, \neg D(x)$. So if the first formula holds on a model, then C does not hold for any element of D does not hold for any element in this model. Hence $\forall x\, (\neg C(x) \vee \neg D(x))$ holds on this model.

7. Functions (3.5 + 4 + 4 points)
We are given the functions $inv\colon \mathbb{R} \to \mathbb{R}$, $dec\colon \mathbb{R} \to \mathbb{R}$, and $exp_2\colon \mathbb{R} \to \mathbb{R}$ defined by

$$inv(x) := \frac{1}{x} \qquad dec(x) := x - 1 \qquad exp_2(x) := 2^x$$

(a) Consider the function f defined by $f := inv \circ inv$. What is the domain of definition D_f of f, and what is its range R_f? Please also give an explicit formula for $f(x)$ (valid for all values of x in the domain of definition).

Solution: The domain of definition and range are $D_f = R_f = \mathbb{R} \setminus \{0\}$. On the domain of definition, $f(x) = x$.

(b) Give an explicit formula for $g(x)$, where g is the function defined by

$$g := inv \circ dec^{-1} \circ exp_2.$$

Solution:

$$g(x) = \frac{1}{2^x + 1}$$

(c) Express the function $h\colon \mathbb{R} \to \mathbb{R}$ defined by

$$h(x) := 2^{\frac{1}{x}} - 2$$

as a composition of the functions inv, dec, exp_2 and their inverses.

Solution:

$$h = exp_2 \circ dec \circ dec \circ inv$$

8. Induction (3 + 8 points)
Consider the sequence $(t_n)_{n=1}^{\infty}$ of integer numbers defined recursively by

$$t_1 := 2, \qquad t_{n+1} := 2 \cdot t_n + 2 - n.$$

We claim that for all $n \geq 1$,

$$t_n = 2^n + n - 1.$$

(a) Verify by explicit calculation that the claim is true for $n = 1$, $n = 2$ and $n = 3$.

Solution:

	t_n	$2 \cdot t_n + 2 - n$	$2^n + n - 1$
$n = 1$	2	$2 \cdot 2 + 2 - 1 = 5$	$2^1 + 1 - 1 = 2 + 0 = 2$
$n = 2$	5	$2 \cdot 5 + 2 - 2 = 10$	$2^2 + 2 - 1 = 4 + 1 = 5$
$n = 3$	10	(irrelevant)	$2^3 + 3 - 1 = 8 + 2 = 10$

So, the statement $t_n = 2^n + n - 1$ is indeed true for $n = 1$, $n = 2$ and $n = 3$.

(b) Prove by mathematical induction that the claim is true for *all* integers $n \geq 1$.

Solution: The base case is covered by part (a), so we now focus on the induction step. Let $m \geq 1$ be arbitrary, and assume that

$$t_m = 2^m + m - 1 \qquad \text{(this is the induction hypothesis).}$$

It then follows that

$$\begin{aligned} t_{m+1} &= 2 \cdot t_m + 2 - m \\ &\overset{\text{IH}}{=} 2 \cdot (2^m + m - 1) + 2 - m \\ &= 2 \cdot 2^m + 2m - 2 + 2 - m \\ &= 2^{m+1} + m \\ &= 2^{m+1} + (m + 1) - 1. \end{aligned}$$

So the induction hypothesis (for $m \geq 1$) implies that the statement

$$t_{m+1} = 2^{m+1} + (m + 1) - 1$$

is true. We conclude that the claim is indeed true for all integers $n \geq 1$.